THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 7: **Measurement of Time and Motion** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If a body doubles its speed, the time to cover the same distance becomes: 1

(a) quarter **(b)** same **(c)** half **(d)** double

2. A line parallel to the time axis at a distance of 30 m shows: 1

(a) the object at rest 30 m away **(b)** speed 0 m/s moving

(c) speed 30 m/s **(d)** motion

3. A car moves at 72 km/h. In 10 s it covers: 1

(a) 20 m **(b)** 720 m **(c)** 200 m **(d)** 72 m

4. If a body travels at the same speed for double the time, the distance becomes: 1

(a) same **(b)** double **(c)** half **(d)** quarter

5. A curve getting steeper in a distance-time graph shows: 1

(a) speed increasing **(b)** rest **(c)** uniform speed **(d)** speed decreasing

6. A pendulum makes 24 oscillations in 1 minute. Its time period is: 1

(a) 24 s **(b)** 2.5 s **(c)** 0.4 s **(d)** 60 s

7. A car's speedometer shows 54 km/h steadily. Its speed in m/s is: 1

(a) 150 **(b)** 15 **(c)** 5.4 **(d)** 54

8. Which graph shape shows an object returning to its starting point? 1

(a) vertical line **(b)** horizontal line

(c) distance rising and then falling **(d)** straight line rising

9. A pendulum has a time period of 1.5 s. In 1 minute it makes: 1

(a) 60 oscillations **(b)** 1.5 oscillations **(c)** 40 oscillations **(d)** 90 oscillations

10. A train 100 m long crosses a pole in 10 s. Its speed is: 1

(a) 1 m/s **(b)** 100 m/s **(c)** 1000 m/s **(d)** 10 m/s

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. A student wants to find the speed of a toy car using a metre scale and stopwatch. Outline the method. 2

12. State the rule for converting km/h to m/s. 2

13. A pendulum takes 50 s for 25 oscillations. Find T and the number of oscillations in 3 min. 2

14. Why do we say 'average speed' for a bus journey? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A car covers the first 60 km at 30 km/h and the next 60 km at 60 km/h. Find the average speed. 3

16. Explain how a water clock can be made at home. 3

17. How does a speedometer differ from the average speed? 3

OR

Two cars A and B: A covers 100 km in 2 h and B covers 150 km in 3 h. Compare their speeds. 3

18. Convert the speeds into m/s and arrange: 72 km/h, 15 m/s, 36 km/h. 3

OR

How can you find your own walking speed? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain how pendulum length changes the time period using data: L = 25 cm (T = 1 s), L = 100 cm (T = 2 s), L = 225 cm (T = 3 s). 5

OR

Explain how electronic timing works in a 100 m race and why it is needed. 5

20. Describe the construction and use of a sundial and a water clock and compare them. 5

OR

Describe an experiment to show that the time period of a pendulum does not depend on the mass of the bob. 5

END OF QUESTION PAPER — Total: 40 Marks

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